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Article Title: Effects of Different Vitamin-C Enriched Collagen Derivatives on Collagen Synthesis

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Running Head: Gelatin, collagen and gummies on collagen synthesis

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"Effects of Different Vitamin-C Enriched Collagen Derivatives on Collagen Synthesis" by Lis DM, Baar K
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Effects of different vitamin-C enriched collagen derivatives on collagen synthesis

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Running Title: Gelatin, collagen and gummies on collagen synthesis

Abbreviations: N-terminal peptide of procollagen I (PINP)

Trial registration: NCT03357432

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Abstract

Nutritional strategies to improve connective tissue collagen synthesis have garnered significant interest, although the scientific validity of these interventions lags behind their hype. This study was designed to determine the effects of three forms of collagen on N-terminal peptide of procollagen I (PINP) and serum amino acid levels. Ten recreationally active males completed a randomized double-blinded cross-over design study consuming either placebo, 15 g of vitamin C enriched gelatin, hydrolyzed collagen or gummy containing equal parts gelatin and hydrolyzed collagen. Supplements were consumed 1 hour before 6 minutes of jump rope. Blood samples were collected immediately prior to supplement consumption and 4 hours after jump rope. A subset of blood samples were collected for amino acid analysis (n=4) 1 hour after ingestion. Consumption of an equivalent dose of each supplement increased amino acids in the circulation similarly across all interventions. PINP levels tended to increase ~20% from baseline in the gelatin and hydrolyzed collagen interventions but not the placebo or gummy. These results suggest that vitamin C enriched gelatin and hydrolyzed collagen supplementation may improve collagen synthesis when taken 1 hour prior to exercise. However, large variability was observed, which precluded significance for any treatment.

Key words: gelatin, collagen, vitamin C, N-terminal peptide of procollagen

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Introduction

Collagen, is the primary structural protein in tendon, ligament, skin and bone (Kjaer, 2004). In athletes, the integrity of these collagen-based tissues is instrumental in force transfer and underpins injury prevention and return to play. Therefore, interventions that improve the robustness of these tissues may have an effect on performance.

Dietary supplementation with gelatin or hydrolyzed collagen (HC) products has become popular among athletes aiming to improve connective tissue function and prevent or treat injury. However, the scientific validity of these nutritional interventions lags behind their hype.

Gelatin and HC have a similar amino acid profile and are particularly high in glycine, proline, hydroxyproline, hydroxylysine and arginine (Eastoe, 1955). Gelatin is isolated from the collagen within the skin, tendons, ligaments and bone of cattle, fish or pigs through boiling. This large molecular weight protein can be further processed using acids or enzymes to produce the small peptides that are termed hydrolyzed collagen. Some variance exists in the peptide sequence and functional properties of the products and this can be influenced both by the source and age of the animal as well as the processes used during extraction (Eastoe, 1955). The main functional difference between the two products is that HC is soluble in water and cannot form a gel.

Gelatin and HC are both digested, absorbed, and incorporated into various connective tissues (Oesser et al., 1999; Shaw et al., 2017). Interestingly, whereas dietary proline and HC are both incorporated into the skin of rats (Matsuda et al., 2006), HC is preferentially incorporated into musculoskeletal tissues including muscle, cartilage, and bone (Oesser et al., 1999). Further, supplementing with HC has been shown to increase cartilage thickness

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(McAlindon et al., 2011) and decrease knee pain in athletes (Clark et al., 2008). However, little is known about how different collagen derivatives differ in digestibility, absorption, or their ability to stimulate collagen synthesis or how collagen synthesis is best measured in a healthy athletic the human body.

Therefore, the current study aimed to determine the effect of different preparations of collagen supplements on collagen synthesis rates. Subjects were provided with 15 g of vitamin C enriched collagen either as a drink containing gelatin or HC or as a gummy containing equal parts gelatin and HC. One hour later, the subjects were asked to jump rope to stimulate bone collagen synthesis and after a further 4 hours, blood was drawn to determine the circulating levels of the N-terminal peptide of procollagen (PINP). PINP in the blood primarily represents bone collagen synthesis (Pollmann 2007). In the current work, it is assumed to be a marker of collagen synthesis in bone and not other collagenous tissues; however, with current methods it is not possible to determine the source of this cleaved peptide in the current study. We hypothesized that collagen synthesis rates would be increased following the consumption of collagen, regardless of whether the collagen was provided as gelatin, hydrolyzed collagen or a gummy.

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Material and Methods

Participants, blood sampling, and exercise intervention

Recreationally active males provided written consent to participate in this repeated measures designed study that was approved by the University of California Davis Institutional Review Board (1140867-1) and was written in accordance with standards set by the Declaration of Helsinki and submitted to the Clinical Trials Registry (ACTRN12616001092482). Prior to commencement of the study, subjects were familiarized

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with the study protocol. On the first day of the study, anthropometric data was collected using a bioelectrical impedance scale (Tanita™ BF-680W) and stadiometer. The subjects then returned to the lab after an overnight fast and provided a baseline blood sample (t= –1 hour) from the antecubital vein of the arm (Figure 1). Immediately following the blood draw, participants ingested 15 g of either gelatin or HC within a diluted cordial beverage (Ribena®, Suntory Limited, UK) or as a gummy containing 7.5 g of both gelatin and HC. Over the course of the study, each subject performed each of the trials in a randomized order separated by at least a 48 hour washout period. The gummy was prepared by heating the water, adding the Ribena®, rapidly mixing in gelatin then the HC into solution. The liquid mixture was then poured into silicon molds, refrigerate and solidified for 24 hours. One hour after consuming the intervention supplement, participants completed 6 minutes of continuous jump rope to load the musculoskeletal system (Shaw et al., 2017). After completing the jump rope, participants were permitted to consume food *ad libitum*, which was recorded and simulated for each subsequent intervention. Participants were also instructed to remain in a rested state with only light activity such as walking or studying permitted. Four hours after jumping rope, a second blood draw was collected (t=4 hour).

Nutritional Supplement

Four isocaloric nutritional supplements were provided to each subject in a random order: (1) 15 g maltodextrin + 80 ml blackcurrant cordial containing 48 mg of vitamin C at time of production (Ribena® Suntory Limited, UK) + 250 ml water (PLA); (2) 15 g gelatin (GreatLakes™, USA) + 80 ml blackcurrant cordial + 250 ml water (GEL); (3) 15 g hydrolyzed collagen (Gelita™, USA) + 80 ml blackcurrant cordial + 250 ml water (HC); and (4); 120 g of

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a solid gummy with ~7.5 g each gelatin and hydrolyzed collagen + blackcurrant cordial + water (GUM). Beverages were mixed well in an opaque drink bottle immediately prior to ingestion and participants were instructed to consume the beverage or food as quickly as possible 1 hour prior to exercise. Treatments were separated by a minimum of 48 hours to avoid carry over effects from the previous treatment. Washout was successful as can be seen in the hydroxylysine levels at baseline in each trial (Figure 3). Nutritional intake in the 12 hours prior to and during each intervention was self-selected by participants under the instruction to manually record and simulate food and beverage consumption for each intervention and to avoid alcohol 24 hours prior to and during each intervention.

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PINP

Blood samples were collected in 3.5 ml clot activated serum separating tubes (Vacutainer Serum Separating Tube, BD, Franklin Lakes, USA), allowed to clot for 20 minutes before centrifugation at 4150 rpm for 10 minutes. Serum samples were then aliquoted into 1.5 ml Eppendorf tubes and stored at -30°C until processing. PINP levels were analyzed by an Enzyme-Linked Immunosorbent Assay (ELISA) kit (Novatein Biosciences, Woburn, MA, USA). Samples were measured in duplicate according the manufacturer instructions using an Epoch microplate spectrophotometer (BioTek, Winooski, USA). The intra-assay %CV was 0.76-1.37% and inter-assay 1.12% and the sensitivity of the assay is 1.0 $pg.ml^{-1}$.

Amino Acid Content of the Blood

A random sub-group (n=4) volunteered to provide a separate aliquot of blood for the purpose of determining the amino acid content of the serum after each treatment. This arm of the study involved a fasted baseline blood sample, ingestion of one of the three

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interventions GEL, HC, GUM in a randomized order on three different occasions separated by 24 hours, then a second blood sample collection 1 hour after supplement ingestion (Figure 1). Whole blood was collected in 3.5 mL clot activated serum separating tubes (BD SST) tubes, allowed to clot for 20 minutes and then centrifuged at 4150 rpm for 10 minutes. The resulting serum was separated into single 1.5 mL microtubes and stored at -30°C until subsequent analysis. Amino acids levels were quantified at the UC Davis Proteomics Core facility by high performance liquid chromatography. Briefly, serum samples were hydrolyzed in 6N HCl for 24 hours at 110°C. 50 µL 10% sulfosalicylic acid was added to 200 µL of the hydrolyzed serum samples. This mixture was diluted 1:2.5 in AE-cysteine buffer, and a 50 µL injection volume was analyzed using a Hitachi L-8900 High Speed Amino Acid Analyzer (Hitachi High-Technologies, Pleasanton, CA).

Statistical Analysis

Sample size calculations for the study were based on the mean changes in PINP and standard deviations established in previous work in this area (Shaw et al., 2017). Given the mean changes and SD in PINP with gelatin supplementation (SD = 12%), a sample size of 8 subjects per group was needed to yield a power of 80% to detect 20% differences with 95% confidence. A ROUT outlier test was performed for analysis of PINP levels to determine outlying data and these outliers were removed from subsequent analysis. PINP levels were analyzed by two-way analysis of variance with one factor being time and the other being the supplement (GraphPad Prism: v.5, San Diego, CA, USA). Data are presented as mean ± standard error of the mean and the significance level was set at $\alpha < 0.05$ for all comparisons. Paired t-test was performed post hoc, to compare each intervention individually to the PLA.

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Amino acid content and PINP levels were analyzed by two-way analysis of variance with one factor being time and the other being the supplement.

Results

The ten recreationally active males who successfully completed the study were 22.7 ± 5.2 years old, 179.0 ± 8.0cm tall, weighed 78.8 ± 7.74kg, and had a body fat of 13.6 ± 3.11%.

Effects of supplementation on PINP

Compared to PLA (-6.8 ± 43.7%), GEL and HC supplementation demonstrated a positive change from baseline PINP 4 hours after completing the jump rope (20.4 ± 52.9 and 19.6 ± 47.5%, respectively) whereas GUM demonstrated a negative change (-23.8 ± 30.5%; Figure 2). Mean changes in PINP from baseline were not statistically significant ($p>0.05$) likely due to the high rate of variability between subjects (five outlying data points were removed).

Amino Acid Concentration in the Blood

We hypothesized that the lower PINP in the GUM group was the result of a decrease in the rate of digestion and absorption of the GUM compared with the GEL and HC groups. To test this hypothesis, subjects were invited back and the circulating levels of amino acids were determined in the blood 1 hour after consuming the supplement. Levels of glycine, lysine, proline, hydroxyproline and hydroxylysine, but not leucine were all elevated at the 1 hour time point. However, in contrast to our hypothesis the increase in each amino acid was consistent across all 3 interventions (Figure 3).

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Discussion

Nutritional interventions, particularly the use of collagen derivatives, have become popular as a means to increase collagen synthesis in the body (Katarzyna & Walczak, 2009). However, there is limited understanding as to how the delivery of these collagen products affect amino acid appearance in the blood and subsequent collagen synthesis. The current study has demonstrated that a dose of 15 g of vitamin C enriched gelatin or hydrolyzed collagen similarly increase PINP, an indirect marker of collagen synthesis. However, likely due to the large variability seen in the control and experimental groups these changes were not statistically significant. Interestingly, a composite of gelatin and hydrolyzed collagen in a vitamin C enriched gummy actually decreased PINP. We hypothesized that the poor response to the gummy was the result of slowed digestion and absorption; however, circulating AA concentrations were similar in all three interventions. Overall, the results from this study suggest that vitamin C enriched gelatin and hydrolyzed collagen ingested prior to 6-minutes of jump rope result in a comparable, albeit modest, increase in collagen synthesis.

The major shortcoming of this study was the extreme variability in the PINP measures in all but the gummy group. This is in stark contrast with our previous work using the exact same ELISA plate (Shaw et al., 2017). A great deal of time was spent troubleshooting the variability of the PINP assay and we discovered that vitamin C can inhibit ELISA reactions (Meng et al., 2005). The ELISA plate manufacturer has since confirmed that adding vitamin C to the samples decreases the colorimetric reaction used for this assay. Several factors may affect vitamin C concertation in the samples and subsequently the ELISA results. Individual variability in subject dietary vitamin C intake as well as the concentration of vitamin C in the

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cordial drink at time of ingestion are the primary factors that could affect vitamin C content of the samples. Vitamin C is known to degrade with storage time, heat and the type of storage container (Igwemmarm et al., 2013; Peleg, 2017). Each bottle or batch of cordial may therefore have very different levels of active vitamin C at time of ingestion. A second important factor that may have contributed to the variation was the duration that the serum was left to clot before centrifugation, pipetting and freezing. The blood processing time may influence vitamin C concentration within the sample. Previously, blood samples were allowed to clot for 2 hours before processing (Shaw et al., 2017), whereas in the current study the blood was centrifuged after 20 minutes of clotting. The longer time may result in the degradation of vitamin C and thus decrease the variability of the resulting ELISA reaction. In support of the hypothesis that difference in vitamin C levels contributed to the variability in the ELISA, the GUM group, where the juice was heated and the vitamin C likely inactivated, showed the least PINP variability of the 4 groups. To further investigate this we have since measured PINP levels in serum allowed to clot for precisely 20 minutes, 1 hour and 2 hours and have shown that variability is least after the 1 hour serum clotting time. Therefore, for groups looking to replicate this work it would be prudent to allow blood samples to clot for exactly 1 hour before processing, regardless if established phlebotomy handling guidelines.

Aside from this potential technical issue this study produced several interesting results. First, the recognized precursors for collagen synthesis, glycine, lysine and proline, and indicators of collagen digestion/absorption (hydroxyproline and hydroxylysine) increased to the same extent in all of the interventions (Figure 3), whereas the key muscle signaling amino acid, leucine, did not. The change in serum amino acids reflects the composition of collagen: glycine (about 33% amino acid residues), proline (12-14%),

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hydroxyproline (<14%) and hydroxylysine (1.5%) (Parry, 1988). However, since the levels of hydroxylysine and hydroxyproline were lowest at baseline, the relative increase in these non-proteinogenic amino acids with feeding was largest. Others have also measured significant increases in hydroxyproline that peak 0.5-2 hours after ingestion (Iwai et al., 2005; Ohara et al., 2007; Shigemura et al., 2011). The changes in glycine, proline, leucine, and lysine following all three feedings were also comparable to those reported by Shaw et al. (Shaw et al., 2017), indicating that the consumption of either GEL, HC, or a gummy results in similar, reliable and repeatable change in serum amino acid levels.

The second interesting finding was that even though all three nutritional interventions increased amino acids similarly, only the GEL and HC treatments tended to increase PINP in the serum. Collagen synthesis increased approximately 20% from baseline in the GEL and HC groups; however, in the GUM or PLA groups there was no increase in PINP 4 hours after completing the 6 minutes of jump rope. The finding that GEL and HC increase PINP ~20%, although not significant, aligns with our previous work demonstrating a 30% increase in serum PINP when consuming 15 g of vitamin C enriched gelatin. In this context, it is interesting that the GUM group did not show an increase in PINP even though serum amino acid levels rose similar to the GEL and HC groups (Figure 3). One possible explanation for this disparity is that in the preparation of the gummies, Ribena® was explosed to heat in order to dissolve the gelatin. Since heating is known to degrade vitamin C (Peleg, 2017), it is possible that following an overnight fast there is not sufficient vitamin C in the serum to support measurable levels of collagen synthesis. Further work is needed to understand the role of vitamin C in the stimulation of collagen synthesis and whether co-ingestion of vitamin C is integral to this process. However, as discussed above, these studies need to employ

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methods to reduce the vitamin C in the serum samples prior to PINP measures in order to minimize the inhibitory effect of vitamin C on the ELISA reaction.

A second discrepancy between the current work and our previous work in this area (Shaw et al., 2017) is that the PLA group did not increase PINP 4 hours after jump rope. We had previously measured an increase in PINP in the placebo (exercise alone) group (Shaw et al., 2017). PINP, is the amino (N) terminal peptide that is cleaved from the procollagen molecule in the process of collagen synthesis and its presence in the serum is a validated marker of bone collagen synthesis (Orum et al., 1996). While the decrease in PINP in the current study might be the result of the vitamin C issue described above, it is also possible that the subjects in the current study were more acclimated to this type of mechanical loading (being boxers and runners) and therefore the loading stimulus would need to be greater in order to stimulate adaptations within the bone (Kjaer, 2004; Shaw et al., 2017).

Another related limitation of the current work is that we don't provide any evidence that the increase in PINP in the blood corresponds with an increase in collagen within tendons, ligaments, or other tissues. In fact, there are cases where changes in PINP within the tendon is opposite the change in collagen incorporation into the same tissue (Hansen et al., 2009) This indicates that collagen synthesis does not always reflect collagen accumulation in a tissue. As a result, further work is needed to determine whether the changes in PINP measured here reflect increased collagen accumulation in tissue.

A third interesting finding in this study was that there was large variability in specific amino acid appearance between subjects and interventions. This biological variability is likely influenced by differences in habitual dietary intake and possibly body weight (Daniel, 2004; Gómez-Guillén et al., 2011; West et al., 2015). Habitual meat intake or polymorphisms

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may directly affect the level or activity of peptide transporters, endopeptidases and/or exopeptidases and influence the absorption of amino acids from various the various collagen derivatives (Daniel, 2004). As the use of collagen supplementation for prophylactic and treatment purposes in sport grows, these are important factors to consider for future investigation of the efficacy of nutritional modulation of collagen synthesis rates (Gómez-Guillén et al., 2011). Another important factor to consider is whether there is anything specifically important about collagen-based proteins for the synthesis of collagen within the body. Put another way, would 15 grams of a different type of protein (whey for example) have the same effect as gelatin or hydrolyzed collagen?

Conclusion

This study demonstrated that ingestion of equivalent doses of gelatin and HC in conjunction with a 6 minute exercise stimulus increased PINP levels comparatively. Large variability in PINP levels were evident which obviated proper statistical analysis between the interventions. However, the fact that the increase in key amino acids was similar for all interventions strengthens the notion that gelatin and hydrolyzed collagen had similar effects on collagen synthesis. One further finding, based on the results from the gummy intervention is that it is possible that co-ingestion of vitamin C may be required to drive collagen synthesis following an overnight fast. Lastly, vitamin C can impair ELISA results and this, alongside a 1 hour blood clotting time, should be considered when replicating the current work.

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Author Contribution

Designed research: KB and DL. Conducted research: DL. Analyzed data and performed statistical analysis: DL, KB. Wrote the paper: DL, KB. Had primary responsibility for final content: KB. All authors read and approved the final version of the manuscript and all authors listed qualify for authorship.

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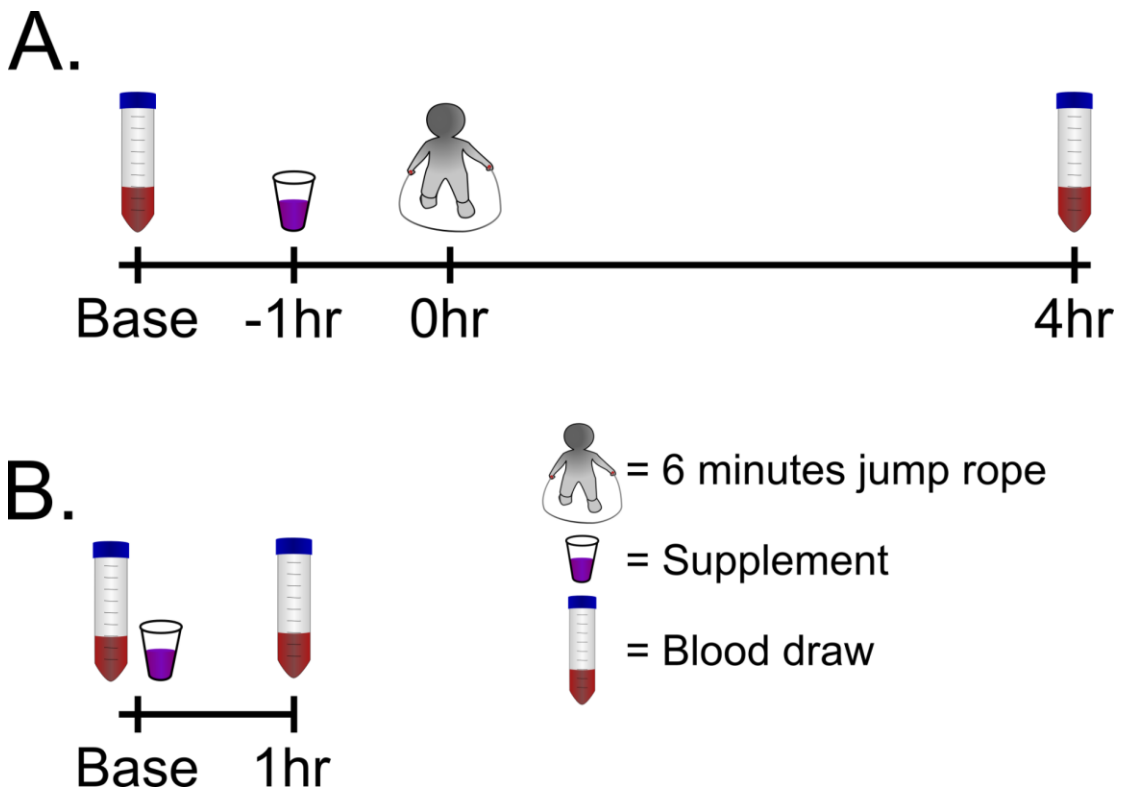

Figure 1: Schematic of study design. (A) Primary study arm (B) Secondary study arm.

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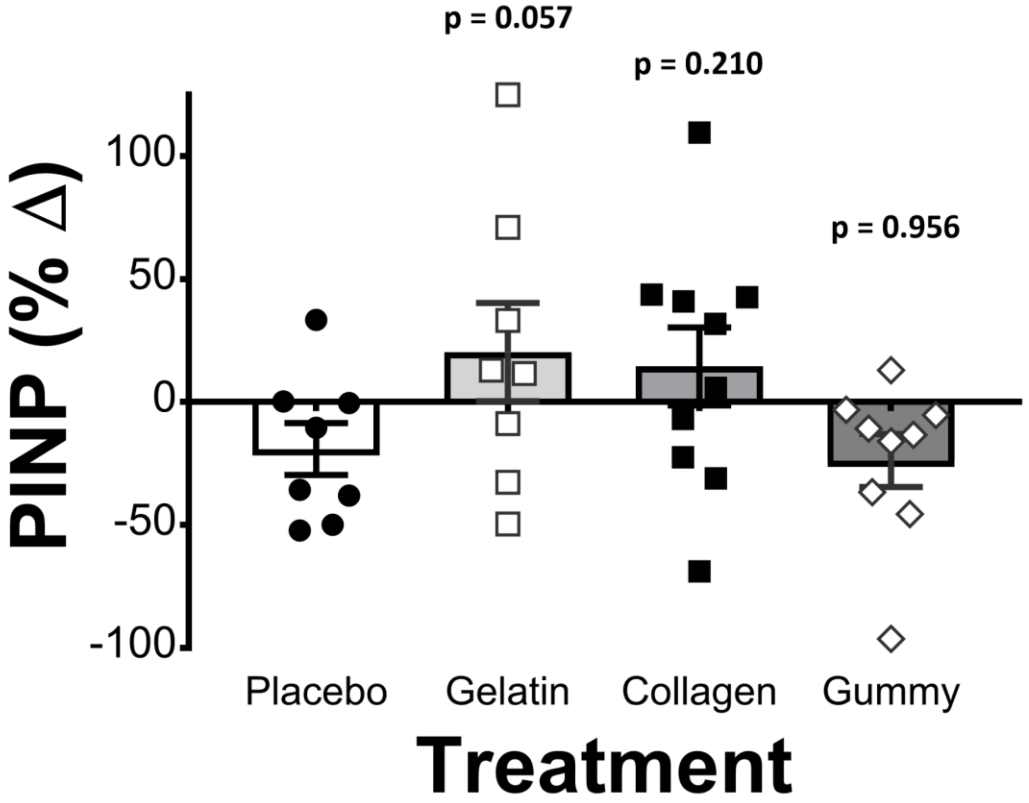

Figure 2: Collagen synthesis following exercise and ingestion of placebo or equivalent dose Vitamin C enriched gelatin, collaen of gummy food, Changes in serum levels of amino-terminal propeptide of collagen (PINP) at 4 hours post exercisxe from baseline (1 hour prior to exercise). Results from paired t-tests between the placebo and each feeding treatment are shown over the appropriate bar. Note the strong trend for significance in the gelatin group. No significance was found between interventions base on the 2-way ANOVA ($p < 0.05$).

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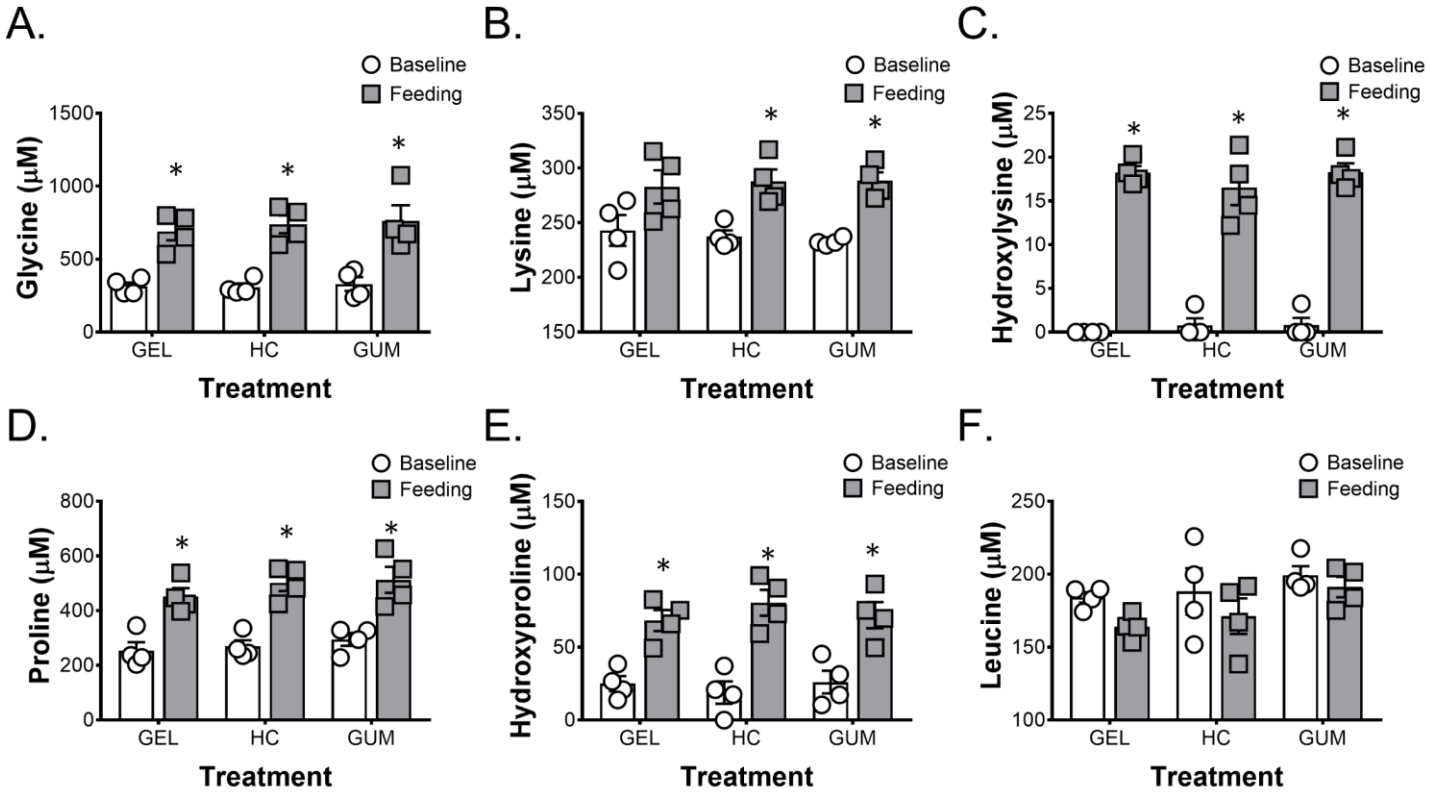

Figure 3: Amino acid concentration in the serum following ingestion of 15 ga of Vitamin C enriched gelatin (GEL), hydrolyzed collagen (HC) or gummy (GUM). The amount of (A) glycine, (B) lysine, (C) hydroxylysine, (D) proline, € hydroxyprolene and (F) leucine in the blood was determined by HPLC by the UC Davis Proteomics Core. The data are presented as mean ± SD for the 4 subjects who completed the cross-over study. The 2-way ANOVA (treatment and time) indicated a significant time effect fro all amino acis tested ($p < 0.05$) except for leucine. There were no significant differences at amino acid levels at baseline and 1 hour post ingestion between any of the interventions.